0.1 Kronecker's Approximation Theorem

Definition 0.1.1. Define a map:

$$[\ \cdot\ ] : \mathbb{R} \to \mathbb{Z} : x \mapsto \max\{n \in \mathbb{Z} \mid x \geq n\}$$

Theorem 0.1.2. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ be given. Then, the set

$$A = \{a + b\alpha \mid a \in \mathbb{N}, b \in \mathbb{Z}\}$$

is dense in $\mathbb{R}$. (That is, $\overline{\mathbb{N} + \mathbb{Z}\alpha} = \mathbb{R}$)

Proof. Observe that: for any $n, m \in \mathbb{N}$ with $n \neq m$,

$$n\alpha - [n\alpha] \neq m\alpha - [m\alpha]$$

Because, if for some distinct $n, m \in \mathbb{N}$ satisfies

$$n\alpha - [n\alpha] = m\alpha - [m\alpha] \implies (n - m)\alpha = [n\alpha] - [m\alpha] \implies \alpha = \frac{[n\alpha] - [m\alpha]}{n - m}$$

, then it implies that α is rational number, contradiction. Now, the map

$$\mathbb{N} \to \mathbb{R} : n \mapsto n\alpha - [n\alpha]$$

is injective. Observe that $\{n\alpha - [n\alpha] \mid n \in \mathbb{N}\} \subseteq [0, 1]$. Build-up is ended. Now, build Pigeon-Holes.
Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $1 < \varepsilon N$. Let N be the smallest such number.
Then, now, there exists N subintervals,

$$\left[0, \frac{1}{N}\right], \left[\frac{1}{N}, \frac{2}{N}\right], \ldots, \left[\frac{N-1}{N}, 1\right]$$

The Pigeonhole Principle gives that: there exists distinct $k, r \in \mathbb{N}$ such that

$$|k\alpha - [k\alpha] - (r\alpha - [r\alpha])| = |[r\alpha] + [k\alpha] + (k - r)\alpha| < \frac{1}{N} < \varepsilon$$

Therefore, for any $\varepsilon > 0$, there exists $p, q \in \mathbb{Z}$ such that $|p + q\alpha| < \varepsilon$. Denote $g = p + q\alpha$. (Note: g means grid!)
Given $x \in \mathbb{R}$, there must exist $k \in \mathbb{Z}$ such that $|x - kg| < \varepsilon$. This implies that the set

$$\{a' + b'\alpha \mid a', b' \in \mathbb{Z}\}$$

is dense in $\mathbb{R}$. (Conti) □